Establishing Appropriate Trust in AI through Transparency and Explainability

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ABSTRACT

As AI systems are increasingly transforming our society, it is critical to support relevant stakeholders to have appropriate understanding and trust in these systems. My dissertation research explores how AI transparency and explainability can help with this goal. I begin with human-centered evaluations of current AI explanation techniques, focusing on their usefulness for people in understanding model behavior and calibrating trust. Next, I identify what explainability needs real AI end-users have and what factors influence their trust through an in-depth case study of a real-world AI application. Finally, I describe two studies, one ongoing and one proposed, that investigate transparency and explainability approaches for Generative AI, such as large language models, to enable safe and successful interactions with this new and powerful technology. My dissertation contributes to both HCI and AI fields by elucidating mechanisms and factors of trust in AI and detailing design considerations for AI transparency and explainability approaches.

CCS CONCEPTS

• **Human-centered computing → Empirical studies in HCI**; • **Computing methodologies → Artificial intelligence**.

KEYWORDS

AI transparency and explainability, Explainable AI, Trust and reliance, Human-AI collaboration

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FOREWORD

I am currently a fourth-year PhD student in the Computer Science department at Princeton University, advised by Professor Olga Russakovsky. I work on AI transparency and explainability to help people better understand and interact with AI systems. My research has been published in both HCI and AI venues (e.g., CHI, FAccT, CVPR, ECCV) and is supported by the NSF Graduate Research Fellowship. I have completed all required coursework and successfully passed the program's general exam in my second year. The expected completion date of my PhD studies is Spring 2025. Upon graduation, I plan to seek research positions where I can continue investigating factors that lead to appropriate trust in AI and advocating for the importance of transparency and explainability.

1 CONTEXT AND MOTIVATION

Appropriate trust is key to safe and successful interactions with AI systems. Despite the rapid growth of technology, AI systems still frequently and unexpectedly fail for various reasons. Users who know when and how much to trust an AI system are likely to effectively use the system and achieve their goals. In contrast, users with unwarranted trust (trusting when the AI is untrustworthy) and unwarranted distrust (distrusting when the AI is trustworthy) are likely to have low-quality interactions, even unsafe ones when they are using the AI in high-stakes settings [1, 17, 36, 49].

Inherently linked to trust in AI are transparency and explainability. Based on information provided by various transparency and explainability approaches, people form an understanding of an AI system's capabilities and limitations, how it works, and how it produced a specific output—and this understanding forms the basis of their trust. Hence, AI transparency and explainability approaches are often viewed as trust calibration methods. Transparency approaches include providing model cards [37], model internals [40], performance measures [23, 27, 43, 51, 52], and (un)certainty information [39, 53]. Explainability approaches include providing local explanations about specific model outputs [13, 16, 24, 50, 54] and global explanations about what the model has learned [2, 3, 15] and how it recognizes a specific class [25, 42, 55]. To date, hundreds of approaches have been proposed, varying in explanation process (e.g., feature attribution [13, 44, 54], counterfactual examples [16, 46]) and form (e.g., heatmap-based [14, 44, 54], part-based [8, 42, 55]).

In contrast to the rapid development of approaches, understandings of when and how AI transparency and explainability lead to (or don't lead to) appropriate understanding and trust fall far behind. I attribute this to the lack of human-centered and context-driven research, and in my dissertation, foreground the people who use AI systems and their needs, goals, and contexts. More concretely, my dissertation makes three key contributions. First, it presents insights from human-centered studies of AI transparency and explainability. Much of existing research focuses on the *technology*, i.e., on developing new techniques and evaluating their technical properties, rather than the *people* who will use or be affected them, or the *context* where they will be deployed. In contrast, my dissertation joins the growing body of work that takes a human-centered

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perspective on AI transparency and explainability [9–11, 28–30]. Through two human evaluation studies [20, 41], it sheds light on how useful proposed approaches are to people, in particular for understanding model behavior and calibrating trust. Then through an in-depth case study of a real-world AI application [21], it surfaces real end-users' explainability needs and goals that are different from those prioritized in current research.

Second, this dissertation provides a more holistic and nuanced understanding of trust in AI. While trust in AI research is fast-growing, there is a lack of empirical studies that approach trust holistically or capture contextual aspects of trust. Most studies are controlled lab experiments that investigate one specific aspect of trust with hypothetical end-users. While they provide valuable insights, studies in real-world contexts are crucial because real end-users' trust relationships with AI may be different from what researchers anticipate. Through a contextually-grounded study [22], this dissertation elaborates on multiple aspects of real end-users trust in AI (e.g., trustworthiness perceptions, trust attitudes, trust-related behaviors) and human, AI, and context-related factors that influence it (see Table 1), expanding the field's understanding of mechanisms and factors of trust.

Finally, this dissertation contributes to the development of transparency and explainability approaches for Generative AI, arguably one of the most influential technologies in the current era. For immediate insights, it investigates an AI transparency approach that can be implemented forthwith for large language models (LLMs) — uncertainty expression through natural language — and examines its effect on user reliance and trust. This study will provide actionable suggestions for when and how LLMs should express uncertainty. For the longer-term, this dissertation proposes a fundamental study of LLMs' self-explanations (i.e., explanations of their own answers and behaviors) that builds on decades of explanations research in psychology and cognitive sciences. The anticipated outcome is a principled framework for explainability in LLMs that will guide the field's future research.

2 RESEARCH QUESTIONS

In summary, my dissertation aims to elucidate mechanisms and factors of trust in AI, and develop AI transparency and explainability approaches that help people form appropriate understanding and trust in AI. To this end, it explores the following research questions grouped into three themes.

(1) Human-centered evaluation of AI explanation techniques
   - RQ1-1: Do current AI explanation techniques help people calibrate their trust? [20]
   - RQ1-2: What are the challenges of using current AI explanation techniques in practice? [41]
(2) Contextually-grounded study of explainability needs and trust in AI
   - RQ2-1: What AI explainability needs do real end-users have, and how do they perceive current explanation approaches? [21]
   - RQ2-2: What factors influence real end-users' trust in AI? [22]
(3) Investigation of transparency and explainability approaches for Generative AI
   - RQ3-1: How does LLMs' natural language expressions of uncertainty affect people's reliance and trust? [*ongoing*]
   - RQ3-2: How do people perceive, evaluate, and act upon LLMs' self-explanations? [*proposed*]

3 RESEARCH METHODS

I use a variety of research methodologies, from running computational experiments with large-scale AI models and datasets to conducting quantitative and qualitative user studies. I choose the specific method based on the purpose of the study. For [20, 41], I implemented, trained, and analyzed numerous AI models and explanation techniques. For [20, 41], I also designed user studies, developed study UIs with HTML and Javascript, and conducted experiments on Amazon Mechanical Turk (MTurk). For [21, 22], I conducted semi-structured interviews and analyzed the gathered data with thematic and abductive coding.

4 FINDINGS TO DATE

4.1 RQ1-1: Do current AI explanation techniques help people calibrate their trust? [20]

The first piece of my dissertation is HIVE [20], a novel evaluation framework for AI explanation techniques. HIVE allows for falsifiable hypothesis testing, cross-method comparison, and human-centered evaluation of AI explanations' ability to help people calibrate their trust in model predictions. My collaborators and I developed it and used it to evaluate four popular techniques (Grad-CAM [44], BagNet [5], ProtoPNet [7], ProtoTree [38]) with 950 participants recruited from MTurk. Notably, we found that participants struggled to distinguish correct and incorrect model predictions based on explanations. Participants also relied more on model predictions, even incorrect ones, when provided explanations. In other words, popular AI explanation techniques engendered over-trust and reliance on AI. This finding issued a warning to the field that AI explanation techniques, even when developed with the best of intentions, can have unintended negative effects in human-AI interaction. The full paper was published at ECCV 2022 [20]. Shorter versions also appeared at CHI 2022's Human-Centered Explainable AI workshop and CVPR 2022's Explainable AI for Computer Vision and Women in Computer Vision workshops.

4.2 RQ1-2: What are the challenges of using current AI explanation techniques in practice? [41]

Continuing the line of work on human-centered evaluation of AI explanations, we next examined factors that affect AI explanations' usefulness in practice. We focused on a class of techniques called concept-based explanations that explain model components and predictions with semantic concepts. They are a particularly promising approach for bridging the gap between complex AI models and human understanding, as they explain AI in units that are intuitive to humans (i.e., semantic concepts). Through an in-depth analysis of four representative techniques (NetDissect [2], TCAV [18], Concept Bottleneck [26], IBD [55]) on multiple datasets (ADE20k [56, 57], Pascal [12], CUB-200-2011 [48]), we identified three commonly

Table 1: Factors of trust in AI identified in our recent work [22]. Different from most existing work, we explored multiple aspects of trust in AI in a real-world context with actual end-users and identified trust-influencing factors in a bottom-up manner. We organized the factors based on whether they are related to the human trustor, the AI trustee, or the context.

Human-related factors	AI-related factors	Context-related factors
Domain knowledge	Ability	Task difficulty
Ability to assess the AI's outputs	Integrity	Perceived risks and benefits
Ability to assess the AI's ability	Benevolence	Situational characteristics
Ability to use the AI	Popularity	Domain's reputation
	Familiarity	Developers' reputation
	Ease of use	

overlooked factors that have a huge effect on explanation quality: (1) the choice of the probe dataset on which explanations are generated, (2) the learnability of concepts in the probe dataset, and (3) the number of concepts used in explanations. We also made immediate suggestions for each factor to improve the usefulness of concept-based explanations. Overall, this work highlights the importance of vetting intuitions when developing and using AI explanation techniques, and equips researchers and practitioners with tools to do so. The full paper was published at CVPR 2023 [41].

4.3 RQ2-1: What AI explainability needs do real end-users have, and how do they perceive current explanation approaches? [21]

The aforementioned works [20, 41] are among the first human-centered evaluations of AI explanation techniques. Still, their evaluation setups lacked *context* because the study participants, i.e., MTurk workers, were not actual end-users of the AI models being explained. Hence, in our next project [21], we interviewed 20 end-users of a widely-used AI application, the Merlin app for bird identification [45], to study what explainability needs end-users have in a real-world context and how they perceive different explanation approaches (heatmap, example, concept, or prototype-based). Intriguingly, we found that participants desired AI explanations for various purposes beyond understanding the AI system, such as learning domain knowledge from the system and giving feedback to developers, expanding the field's understanding of explainability needs. We also found that participants' perceptions of different explanation approaches vary with respect to their domain and AI knowledge base. These findings provide insights into the nuances of real-world human-AI interactions and highlight that human-centered and contextually-grounded research is necessary to develop effective AI transparency and explainability approaches. The full paper was published at CHI 2023 [21] and received a best paper honorable mention award. Shorter versions also appeared at NeurIPS 2022's Human-Centered AI workshop and CHI 2023's Human-Centered Explainable AI workshop.

4.4 RQ2-2: What factors influence real end-users' trust in AI? [22]

In the same interviews, we also inquired about participants' trust in the AI application from many angles with questions about their typical use of the app, as well as whether they would use the app in hypothetical higher-risk scenarios. We analyzed this data in a separate paper [22]. Different from most prior work, which investigates one aspect of trust, we analyzed multiple aspects of trust based on the seminal trust model by Mayer et al. [34] that delineates trust from its antecedents, context, and products. Our holistic approach to trust revealed a comprehensive picture of end-users' trust relationships with AI that cannot be gained by studying only one aspect of trust. Notably, we found a discrepancy between participants' *general* trustworthiness perceptions and trust attitudes, and *instance-specific* trust-related behaviors, adding nuances to existing understandings of trust in AI. Our bottom-up study approach also allowed us to identify a wide range of trust-influencing factors, organized in Table 1 based on whether they are related to the human trustor, the AI trustee, or the context. This work significantly deepens the field's understanding of mechanisms and factors of trust, and yield insights into how readily existing theories of trust (e.g., Mayer et al.'s trust model [34]) can be operationalized for empirical research. The full paper was published at FAccT 2023 [22]. A shorter version appeared at CHI 2023's Trust and Reliance in AI-assisted Tasks workshop.

5 EXPECTED NEXT STEPS

5.1 RQ3-1: How does LLMs' natural language expressions of uncertainty affect people's reliance and trust? [19]

Based on the insights from my prior work, I am currently investigating transparency and explainability approaches for Generative AI that is having a rapidly growing impact on our society. As a first step, I have been examining the impact of LLMs' uncertainty expression, a form of transparency [4], on user reliance and trust. This work began in Summer 2023 during my internship at Microsoft Research in the FATE (Fairness, Accountability, Transparency, and Ethics in AI) group. The goal of this work is to understand whether LLMs' natural language expressions of uncertainty can reduce over-reliance and trust, a well-known pitfall that has been shown to reduce task performance and worsen user experience [6, 47]. We explore this question in the context of LLM-infused search engines such as Copilot in Bing as they are already used by millions of people, and because search is a domain where the factual correctness of answers is fundamental. Over the past months, we carefully designed and ran a large-scale, pre-registered, human-subject experiment (N=404). Our findings suggest that using natural language

expressions of uncertainty may be an effective approach for reducing over-reliance and trust on LLMs, but that the precise language used matters: expressions from a first-person perspective (e.g., "*I'm not sure, but…*") were more effective than expressions from a general perspective (e.g., "*There is uncertainty, but…*") in our experiment. We anticipate our findings will inform the design of a wide range of LLM-infused applications. The paper is currently in submission.

5.2 RQ3-2: How do people perceive, evaluate, and act upon LLMs' self-explanations? [*proposed*]

The final piece of my dissertation will be an investigation of LLMs' self-explanations. Different from other types of AI models (e.g., classification models), LLMs can and often provide explanations for their answers and behaviors (e.g., "*The answer to your question is X because…*", "*I cannot handle your request because…*"), even when explanations were not requested. I am particularly interested in these self-explanations because they can be highly convincing and fluent, while lacking faithfulness, relevance, and other desirable properties of explanations. They are likely already impacting millions of users, but very little is known about how people perceive, evaluate, and act upon them. To tackle this problem, I will first develop a taxonomy of properties for LLMs' self-explanations, building on the rich literature on explanations from psychology and cognitive sciences [31, 32, 35]. With this taxonomy, I will then investigate how different explanation properties affect people's satisfaction and trust in LLMs' self-explanations. For example, prior research in psychology has found that people prefer selective explanations [33]. I am curious if this finding will still hold for LLMs and whether there are other more important explanation properties. The anticipated outcome of this work is a principled framework for explainability in LLMs, that can guide the field's future research and design of LLMs' explanations.

6 RESEARCH COMMUNITY ACTIVITIES

Beyond my research work, I am passionate about building and connecting research communities. This past year, I helped build a community for Explainable AI (XAI) researchers and practitioners by co-managing the ExplainableAIWorld slack group (380+ people) and the @XAI_Research twitter account (1600+ followers). I also contributed to organizing a talk series for junior researchers to share their work and meet other researchers. In June 2023, I led the organization of the Explainable AI for Computer Vision (XAI4CV) workshop at CVPR 2023. In organizing the workshop, I made active efforts to bridge the HCI and AI research communities by inviting distinguished researchers in both fields as keynote speakers, and creating a new demo track to emphasize the role of communication and interaction in explainability research. The workshop was hugely successful with over 200 people in attendance. Currently, I am part of two workshop organizing committees. One is for another iteration of the XAI4CV workshop at CVPR 2024. Another is for the Human-Centered Explainable AI (HCXAI) workshop at CHI 2024. I hope these workshops further connect the HCI and AI research communities and encourage diverse perspectives and approaches to AI transparency and explainability.

7 CONCLUSION

Appropriate trust is key to safe and successful interactions with AI systems. My dissertation aims to elucidate mechanisms and factors of trust in AI and develop AI transparency and explainability approaches that help people form appropriate understanding and trust. As a first step, I conducted human-centered evaluations of current AI explanation techniques' ability to help people understand model behavior and calibrate their trust. I then conducted an in-depth case study of a real-world AI application and identified actual end-users' explainability needs and trust relationship with AI. As next steps, I described two studies, one ongoing and one proposed, that investigate transparency and explainability approaches for Generative AI that is having a transformative impact on our society. Together, this dissertation contributes to both HCI and AI fields and lays out actionable steps for establishing appropriate understanding and trust in AI.

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